Kirchenfeld Robotics

CABLE GUARD

Sponsored by

OUR SOLUTION

AUTONOMOUS ON-ROPE INSPECTION

Driven Wheel
TPU groove grips on rope

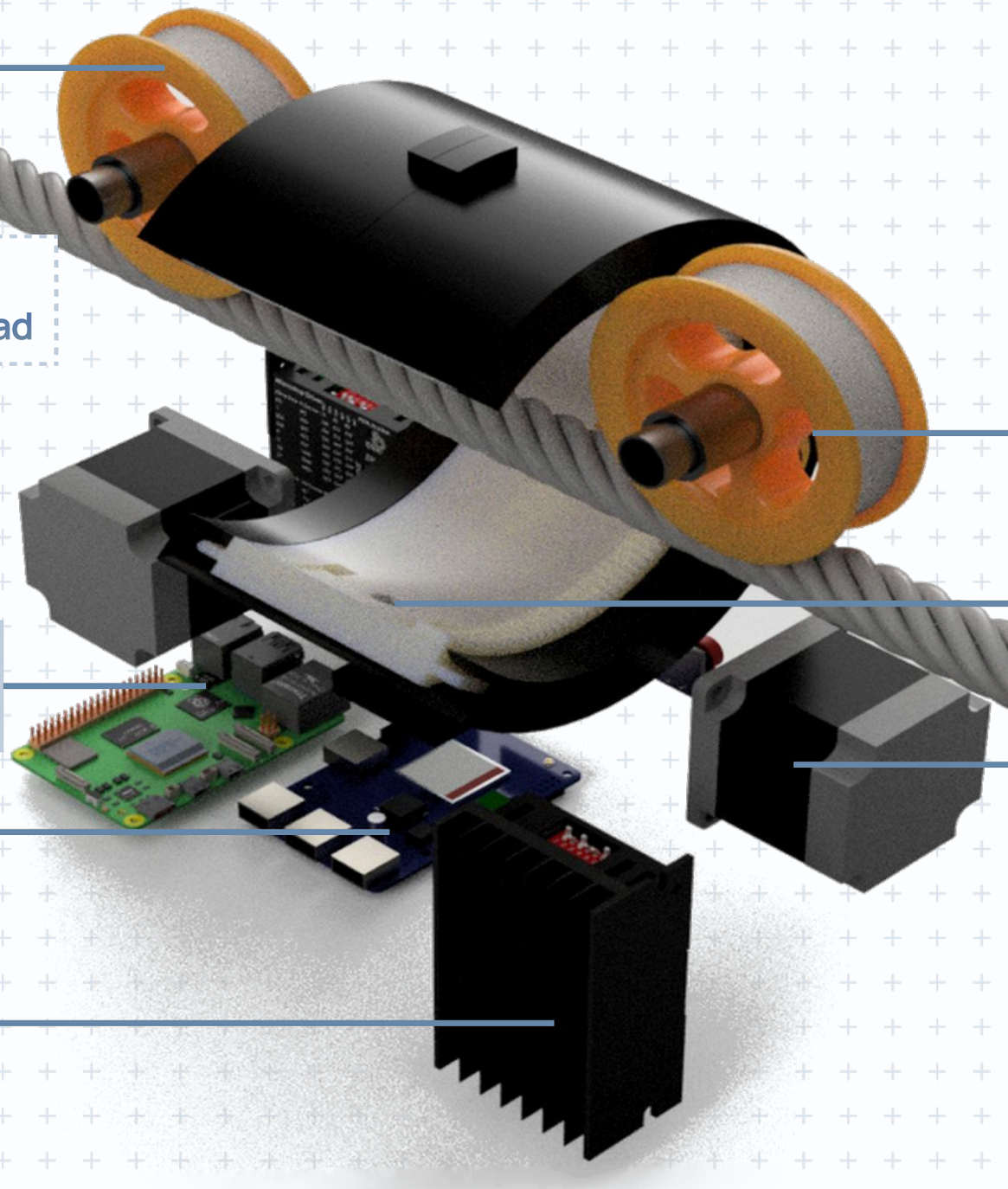
Laser Sensor
sees rope sockets 3cm ahead

Controls
buttons, buzzer & display.

Raspberry Pi 5
runs the onboard AI model

LTE Module
sends findings to server

2x TB6600
motor drivers



Size
277 × 160 × 245 mm

Cost
≈ CHF 400 for the prototype

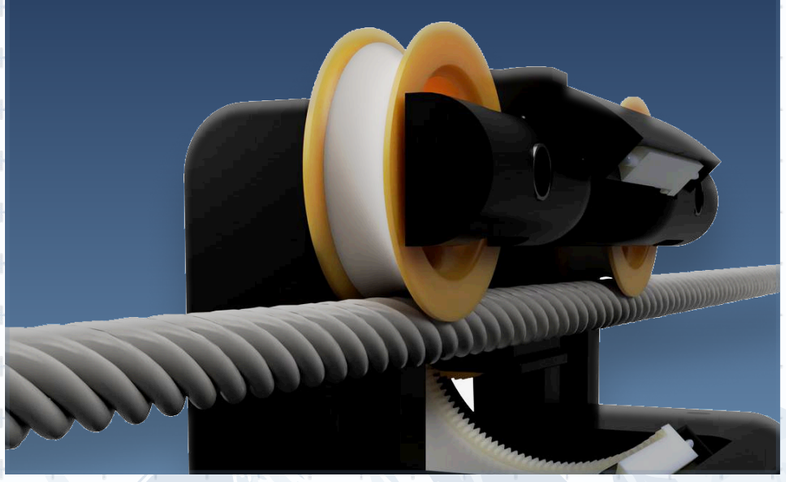
Free Wheel

Camera Ring
2 cameras, 90° opening

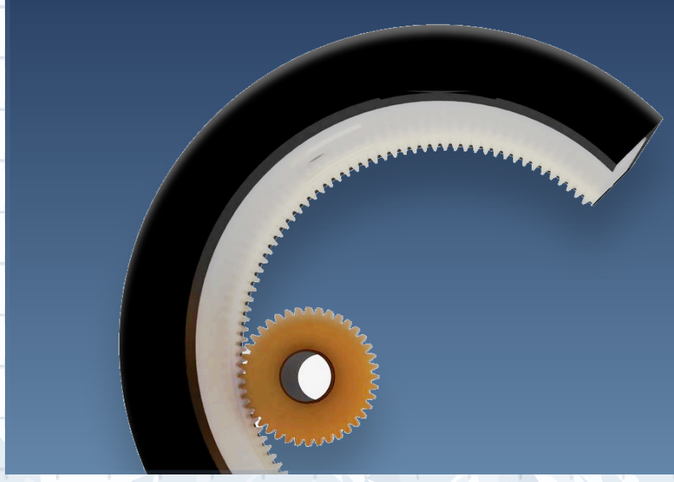
2x NEMA 23
drive + ring motors

PROGRESS

Wheels with Grip



Camera Ring



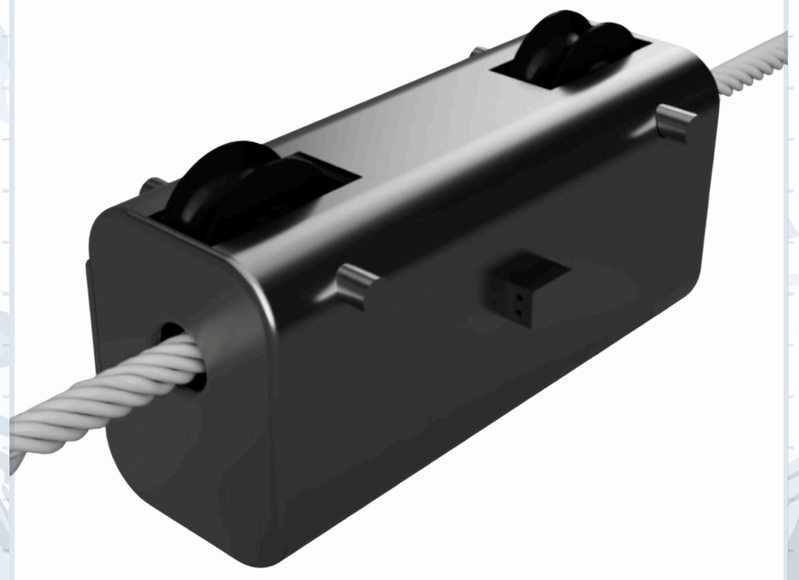
Ring Drive



V1, MAY
1 camera, AI on a server



V2, JUNE
8 Hall sensors



V3, OCTOBER
Camera Ring, Onboard AI

